

WI4014TU – Numerical Analysis for PDE's

Lecture 5

Finite Difference Method in 2D

Properties of the 2D FD Laplacian matrix

Finite Difference Method in 2D

Poisson's equation in 2D

Let $\Omega \subset \mathbb{R}^2$ be a rectangle $(a, b) \times (c, d)$ with the boundary $\partial\Omega$. Consider the problem

$$\begin{aligned} -\Delta u(x, y) &= f(x, y), \quad (x, y) \in \Omega \\ u(x, y) &= 0, \quad (x, y) \in \partial\Omega \end{aligned}$$

where

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

2D grid

2D grid on $\Omega = (a, b) \times (c, d)$ is built from two uniform 1D grids

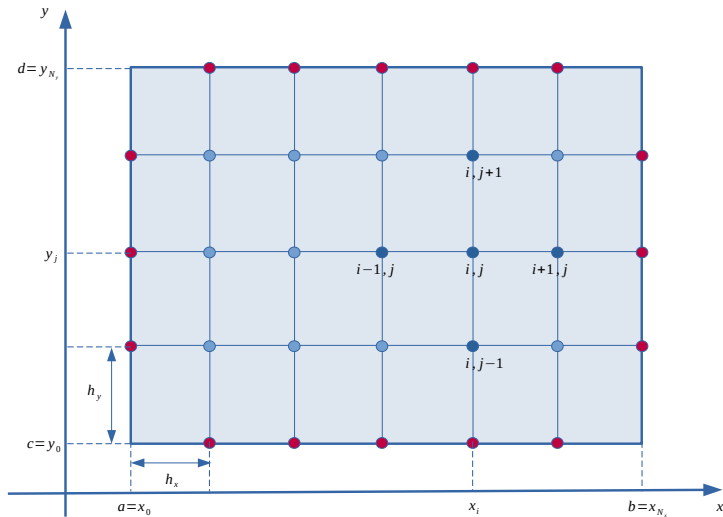
$$\Omega_h = \{(x_i, y_j) \mid x_i = a + ih_x, i = 0, 1, \dots, N_x; y_j = c + jh_y, j = 0, 1, \dots, N_y\}$$

where

$$h_x = \frac{b - a}{N_x}, \quad h_y = \frac{d - c}{N_y}$$

Grid values are $u_{i,j} = u(x_i, y_j)$ and $f_{i,j} = f(x_i, y_j)$.

2D grid



FD approximation of negative 2D Laplacian

$$-\frac{\partial^2 u_{i,j}}{\partial x^2} = \frac{-u_{i-1,j} + 2u_{i,j} - u_{i+1,j}}{h_x^2} + \mathcal{O}(h_x^2)$$

$$-\frac{\partial^2 u_{i,j}}{\partial y^2} = \frac{-u_{i,j-1} + 2u_{i,j} - u_{i,j+1}}{h_y^2} + \mathcal{O}(h_y^2)$$

$$\begin{aligned} -\Delta u_{i,j} &= \frac{-u_{i-1,j} + 2u_{i,j} - u_{i+1,j}}{h_x^2} + \frac{-u_{i,j-1} + 2u_{i,j} - u_{i,j+1}}{h_y^2} \\ &\quad + \mathcal{O}(h_x^2) + \mathcal{O}(h_y^2) \end{aligned}$$

FD discretization of 2D Poisson's equation

Inner points:

$$\frac{-u_{i-1,j} + 2u_{i,j} - u_{i+1,j}}{h_x^2} + \frac{-u_{i,j-1} + 2u_{i,j} - u_{i,j+1}}{h_y^2} = f_{i,j}$$
$$i = 1, 2, \dots, N_x - 1, \quad j = 1, 2, \dots, N_y - 1$$

Boundary points (homogeneous Dirichlet BC's, $u(x, y) = 0$, $(x, y) \in \partial\Omega$):

$$u_{i,j} = 0, \quad \text{if}$$

lower (South) boundary: $i = 0, 1, \dots, N_x$; $j = 0$;

upper (North) boundary: $i = 0, 1, \dots, N_x$; $j = N_y$;

left (West) boundary: $i = 0$; $j = 0, 1, \dots, N_y$;

right (East) boundary: $i = N_x$; $j = 0, 1, \dots, N_y$;

Sorting the unknowns: lexicographic order

One way to store all unknown grid values $u_{i,j}$ in a single vector

$\mathbf{u} \in \mathbb{R}^{(N_x-1)(N_y-1)}$ is to sort them in **lexicographic order** - first change x , then change y . Run through all x_i for y_1 , run through all x_i for y_2 , etc:

$$\mathbf{u} = \begin{bmatrix} u_{1,1} \\ u_{2,1} \\ \vdots \\ u_{N_x-1,1} \\ u_{1,2} \\ u_{2,2} \\ \vdots \\ u_{N_x-1,2} \\ \vdots \\ u_{1,N_y-1} \\ u_{2,N_y-1} \\ \vdots \\ u_{N_x-1,N_y-1} \end{bmatrix}$$

Warning: size of FD Laplacian in 2D

- The Laplacian matrix A can be very large. For example, if $N_x = N_y = 1001$, then $\mathbf{u} \in \mathbb{R}^{10^6}$ has 10^6 elements, while $A \in \mathbb{R}^{10^6 \times 10^6}$ has 10^{12} elements
- If all elements are stored as double precision floating point numbers, 8 bytes each, then $\mathbf{u}$ takes 8×10^6 bytes, i.e., 8 MB, whereas A would take 8×10^{12} bytes, i.e., 8 TB.
- Fortunately, most of the elements of A are zeros, i.e., A is **sparse**. There are special economic ways for storing sparse matrices and working with them
- While in *compiled* languages (C/C++, FORTRAN) the 2D Laplacian matrix may be constructed using nested loops, in *interpreted* languages (Matlab, Python) nested loops can be very slow and special ‘vectorized’ ways for constructing matrices are used

An efficient way to construct 1D Laplacian

The 1D Laplacian matrix $A \in \mathbb{R}^{(N-1) \times (N-1)}$

$$A = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix}$$

can be obtained from the sparse matrix $D \in \mathbb{R}^{N \times (N-1)}$ of the backward first-order derivative:

$$D = \frac{1}{h} \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & 0 & \dots & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

simply as the (sparse) matrix-matrix product

$$A = D^T D$$

Partial Laplacian matrices

Let $D_x \in \mathbb{R}^{N_x \times (N_x-1)}$ and $D_y \in \mathbb{R}^{N_y \times (N_y-1)}$ be the sparse first-order (partial) derivative matrices

$$D_x = \frac{1}{h_x} \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & 0 & \dots & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}, \quad D_y = \frac{1}{h_y} \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ -1 & 1 & 0 & 0 & \dots & 0 \\ \vdots & & & & & \vdots \\ 0 & & & & & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

Define two sparse (partial) 1D Laplacian matrices as

$$A_{xx} = D_x^T D_x \quad \text{and} \quad A_{yy} = D_y^T D_y$$

with $A_{xx} \in \mathbb{R}^{(N_x-1) \times (N_x-1)}$, $A_{yy} \in \mathbb{R}^{(N_y-1) \times (N_y-1)}$

Sparse 2D Laplacian matrix via Kronecker products

Introducing the identity matrices $I_x = I_{N_x-1}$ and $I_y = I_{N_y-1}$ we can construct the sparse 2D Laplacian matrix A corresponding to the lexicographic order of unknowns as

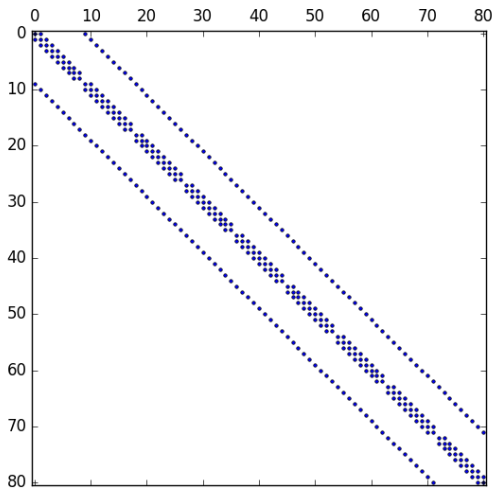
$$A = I_y \otimes A_{xx} + A_{yy} \otimes I_x$$

where $\otimes$ denotes the Kronecker product.

Given any two matrices, e.g. $A \in \mathbb{R}^{2 \times 3}$ and $B \in \mathbb{R}^{m \times n}$, the **Kronecker product** $A \otimes B \in \mathbb{R}^{2m \times 3n}$ is defined as:

$$A \otimes B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & a_{13}B \\ a_{21}B & a_{22}B & a_{23}B \end{bmatrix}$$

Structure of 2D Laplacian matrix in lexicographic order



Properties of 2D FD (negative) Laplacian matrix

Symmetry

Theorem: The 2D FD negative Laplacian matrix A (with Dirichlet BC's on a rectangular domain) is symmetric.

Proof. Symmetry means that $A^T = A$. We are going to show that $\mathbf{v}^T A \mathbf{u} = (\mathbf{v}^T A \mathbf{u})^T = \mathbf{u}^T A^T \mathbf{v} = \mathbf{u}^T A \mathbf{v}$ for any $\mathbf{u}$ and $\mathbf{v}$ that satisfy zero Dirichlet BC's on $\partial\Omega$. We consider a doubly-uniform grid $h_x = h_y = h$. Proof for uniform grid is similar. First, we rewrite the inner point equation as:

$$\begin{aligned} & \frac{-u_{i-1,j} - u_{i,j-1} + 4u_{i,j} - u_{i,j+1} - u_{i+1,j}}{h^2} \\ &= \frac{u_{i,j} - u_{i-1,j}}{h^2} - \frac{u_{i+1,j} - u_{i,j}}{h^2} + \frac{u_{i,j} - u_{i,j-1}}{h^2} - \frac{u_{i,j+1} - u_{i,j}}{h^2} \\ &= - \left(\frac{\mathcal{D}_x^+ u_{i,j} - \mathcal{D}_x^- u_{i,j}}{h} + \frac{\mathcal{D}_y^+ u_{i,j} - \mathcal{D}_y^- u_{i,j}}{h} \right) \end{aligned}$$

Proof of symmetry (continued)

$$\begin{aligned}\mathbf{v}^T A \mathbf{u} &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} v_{i,j} \frac{-u_{i-1,j} - u_{i,j-1} + 4u_{i,j} - u_{i,j+1} - u_{i+1,j}}{h^2} \\ &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} v_{i,j} \left(\frac{u_{i,j} - u_{i-1,j}}{h^2} - \frac{u_{i+1,j} - u_{i,j}}{h^2} \right. \\ &\quad \left. + \frac{u_{i,j} - u_{i,j-1}}{h^2} - \frac{u_{i,j+1} - u_{i,j}}{h^2} \right)\end{aligned}$$

Proof of symmetry (continued)

For any fixed $j = 1, \dots, N_y - 1$ the following holds:

$$\begin{aligned} & \sum_{i=1}^{N_x-1} v_{i,j} \left(\frac{u_{i,j} - u_{i-1,j}}{h^2} - \frac{u_{i+1,j} - u_{i,j}}{h^2} \right) \\ &= \sum_{i=1}^{N_x-1} v_{i,j} \frac{u_{i,j} - u_{i-1,j}}{h^2} - \sum_{i=1}^{N_x-1} v_{i,j} \frac{u_{i+1,j} - u_{i,j}}{h^2} \\ &= \sum_{i=1}^{N_x-1} v_{i,j} \frac{u_{i,j} - u_{i-1,j}}{h^2} - \sum_{i=2}^{N_x} v_{i-1,j} \frac{u_{i,j} - u_{i-1,j}}{h^2} \\ &= \sum_{i=1}^{N_x} v_{i,j} \frac{u_{i,j} - u_{i-1,j}}{h^2} - \sum_{i=1}^{N_x} v_{i-1,j} \frac{u_{i,j} - u_{i-1,j}}{h^2} \\ &= \sum_{i=1}^{N_x} \frac{v_{i,j} - v_{i-1,j}}{h} \frac{u_{i,j} - u_{i-1,j}}{h} \end{aligned}$$

Proof of symmetry (continued)

Similarly, for any fixed $i = 1, \dots, N_x - 1$ the following holds:

$$\begin{aligned} & \sum_{j=1}^{N_y-1} v_{i,j} \left(\frac{u_{i,j} - u_{i,j-1}}{h^2} - \frac{u_{i,j+1} - u_{i,j}}{h^2} \right) \\ &= \sum_{j=1}^{N_y} \frac{v_{i,j} - v_{i,j-1}}{h} \frac{u_{i,j} - u_{i,j-1}}{h} \end{aligned}$$

Proof of symmetry (continued)

$$\begin{aligned}\mathbf{v}^T \mathbf{A} \mathbf{u} &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} v_{i,j} \left(\frac{u_{i,j} - u_{i-1,j}}{h^2} - \frac{u_{i+1,j} - u_{i,j}}{h^2} \right. \\ &\quad \left. + \frac{u_{i,j} - u_{i,j-1}}{h^2} - \frac{u_{i,j+1} - u_{i,j}}{h^2} \right) \\ &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x} \frac{v_{i,j} - v_{i-1,j}}{h} \frac{u_{i,j} - u_{i-1,j}}{h} + \sum_{j=1}^{N_y} \sum_{i=1}^{N_x-1} \frac{v_{i,j} - v_{i,j-1}}{h} \frac{u_{i,j} - u_{i,j-1}}{h}\end{aligned}$$

These expressions are symmetric in $u_{i,j}$ and $v_{i,j}$. Hence, $\mathbf{v}^T \mathbf{A} \mathbf{u} = \mathbf{u}^T \mathbf{A} \mathbf{v}$.

Positivity

Theorem: The 2D FD negative Laplacian matrix A (with Dirichlet BC's on a rectangular domain) is positive-definite.

Proof. We are going to show that $\mathbf{u}^T A \mathbf{u} > 0$ for any $\mathbf{u} \neq \mathbf{0}$. Using the inner point equation in the symmetrized form:

$$\begin{aligned}\mathbf{u}^T A \mathbf{u} &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x} \left(\frac{u_{i,j} - u_{i-1,j}}{h^2} \right)^2 + \sum_{j=1}^{N_y} \sum_{i=1}^{N_x-1} \left(\frac{u_{i,j} - u_{i,j-1}}{h^2} \right)^2 \\&= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} \left(\frac{u_{i,j} - u_{i-1,j}}{h^2} \right)^2 + \sum_{j=1}^{N_y-1} \left(\frac{u_{N_x,j} - u_{N_x-1,j}}{h^2} \right)^2 \\&\quad + \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} \left(\frac{u_{i,j} - u_{i,j-1}}{h^2} \right)^2 + \sum_{i=1}^{N_x-1} \left(\frac{u_{i,N_y} - u_{i,N_y-1}}{h^2} \right)^2 > 0\end{aligned}$$

Eigenvalues and eigenvectors on a rectangle domain

Eigenproblem for FD Laplacian on a doubly uniform grid: $h_x = h_y = h$ with zero Dirichlet B.C.'s on $\partial\Omega$. Matrix form: $A\mathbf{v}_k = \lambda_k \mathbf{v}_k$ can be written as the system of equations:

$$\begin{aligned} \frac{1}{h^2} [-v_k(i-1, j) - v_k(i, j-1) + 4v_k(i, j) - v_k(i+1, j) - v_k(i, j+1)] \\ = \lambda_k v_k(i, j); \quad i = 1, 2, \dots, N_x - 1, j = 1, 2, \dots, N_y - 1 \\ v_k(0, :) = v_k(N_x, :) = v_k(:, 0) = v_k(:, N_y) = 0, \quad \mathbf{v}_k \neq 0 \end{aligned}$$

consider the (candidate) eigenvector $\mathbf{v}_k$ with the elements:

$$v_{k_x, k_y}(i, j) = \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right)$$

Using $\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$ and $1 - \cos(2\alpha) = 2 \sin^2(\alpha)$.

$$\begin{aligned}
& \frac{1}{h^2} \left[-\sin\left(\frac{\pi(i-1)k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) - \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi(j-1)k_y}{N_y}\right) \right. \\
& \quad + 4 \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) - \sin\left(\frac{\pi(i+1)k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) \\
& \quad \left. - \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi(j+1)k_y}{N_y}\right) \right] \\
&= \frac{1}{h^2} \left[-2 \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) \cos\left(\frac{\pi k_x}{N_x}\right) - 2 \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) \cos\left(\frac{\pi k_y}{N_y}\right) \right. \\
& \quad \left. + 4 \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) \right] \\
&= \frac{2}{h^2} \left[2 - \cos\left(\frac{\pi k_x}{N_x}\right) - \cos\left(\frac{\pi k_y}{N_y}\right) \right] \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right) \\
&= \frac{4}{h^2} \left[\sin^2\left(\frac{\pi k_x}{2N_x}\right) + \sin^2\left(\frac{\pi k_y}{2N_y}\right) \right] \sin\left(\frac{\pi i k_x}{N_x}\right) \sin\left(\frac{\pi j k_y}{N_y}\right)
\end{aligned}$$

$$\lambda_{k_x, k_y} = \frac{4}{h^2} \left[\sin^2 \left(\frac{\pi k_x}{2N_x} \right) + \sin^2 \left(\frac{\pi k_y}{2N_y} \right) \right]$$

$$v_{k_x, k_y}(i, j) = \sin \left(\frac{\pi i k_x}{N_x} \right) \sin \left(\frac{\pi j k_y}{N_y} \right)$$

$$i = 1, \dots, N_x - 1, \quad j = 1, \dots, N_y - 1.$$

$$k_x = 1, \dots, N_x - 1, \quad k_y = 1, \dots, N_y - 1.$$

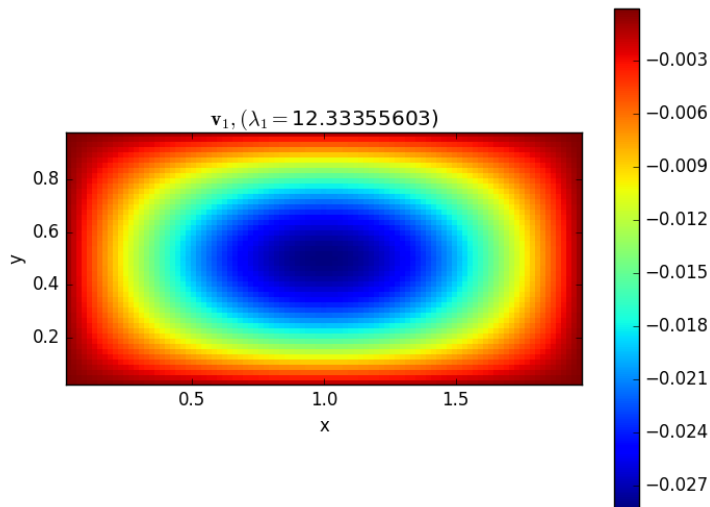
Hence, on rectangular domains the matrix is:

- Symmetric $A^T = A$
- Positive-definite, since $\lambda_{\min} > 0$

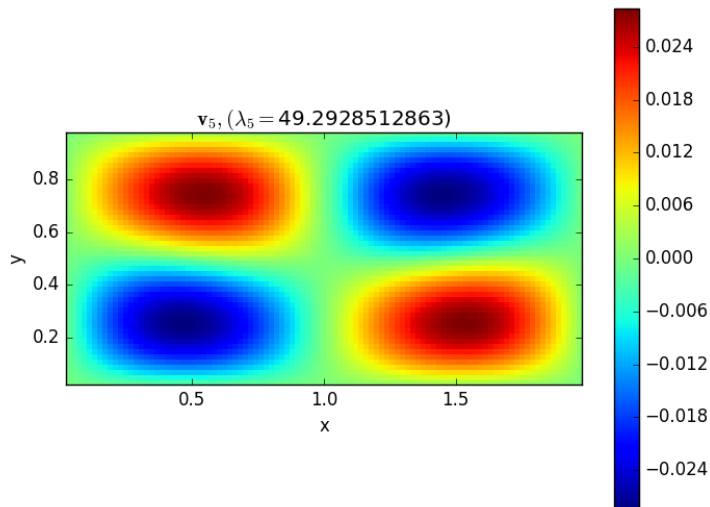
Computing a few eigenvalues numerically

- It is not possible to find eigenvalues of very large matrices numerically
- There exist methods to find a few selected eigenvalues, e.g., 10 eigenvalues with the largest amplitude (easiest), or with the smallest amplitude (difficult), or closest to some point in complex plane (difficult)
- This algorithm is implemented in the function `scipy.sparse.linalg.eigs()`
- There is a special version for Hermitian matrices: `scipy.sparse.linalg.eigsh()`

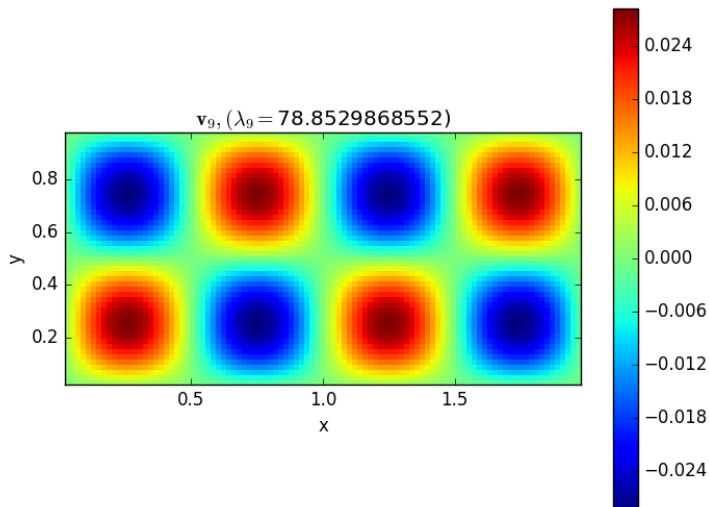
1-st eigen-pair



5-th eigen-pair



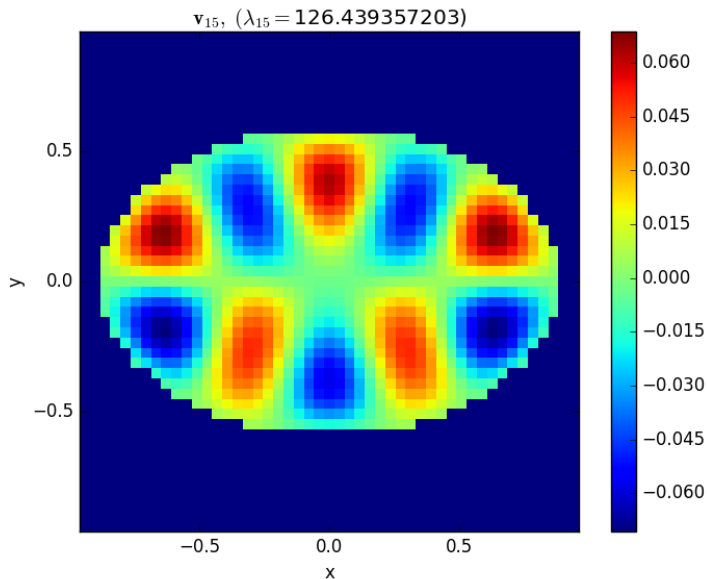
9-th eigen-pair



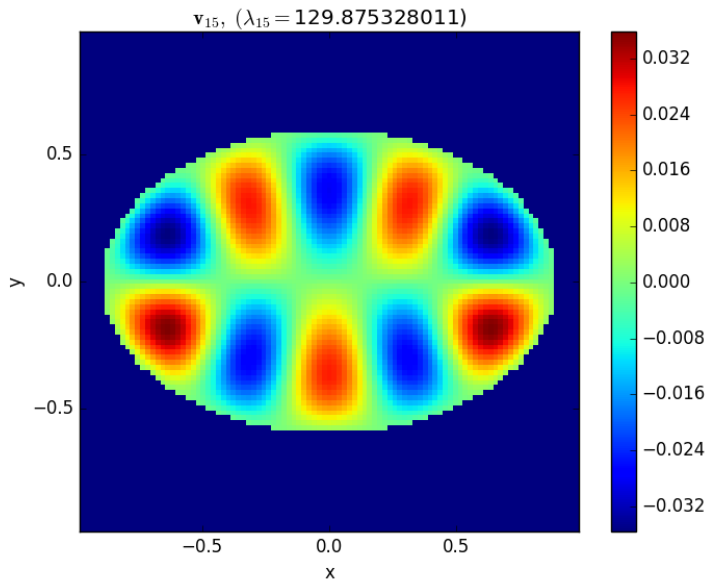
Non-rectangular domains: numerical eigenvectors

- If you are OK with the geometric error in the representation of non-rectangular boundaries (staircase effect), then FD can be used to model domains of any shape.
- However, the eigenvalues and eigenvectors can not be obtained exactly for general domains.

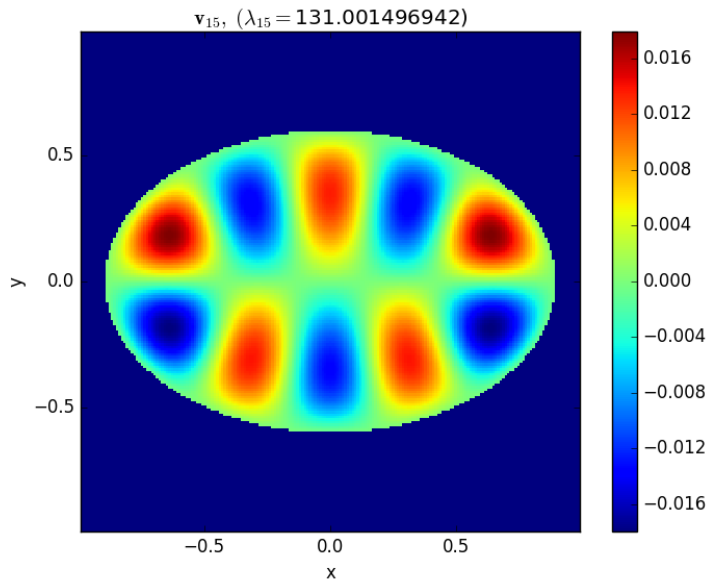
15-th eigen-pair: rough grid



15-th eigen-pair: finer grid



15-th eigen-pair: even finer grid



Diagonal dominance

A is diagonally dominant, i.e., $|a_{nn}| \geq \sum_{m \neq n} |a_{nm}|$:

inner-point row of A : $\left[\cdots -\frac{1}{h^2} \cdots -\frac{1}{h^2} \frac{4}{h^2} -\frac{1}{h^2} \cdots -\frac{1}{h^2} \cdots \right]$

Rows of A that correspond to boundary points are strictly diagonally dominant, i.e., $|a_{nn}| > \sum_{m \neq n} |a_{nm}|$

boundary-point row of A : $\left[\cdots -\frac{1}{h^2} \frac{4}{h^2} -\frac{1}{h^2} \cdots -\frac{1}{h^2} \cdots \right]$

A is a Z matrix

A is a Z matrix, i.e., $a_{nm} \leq 0$, $n \neq m$:

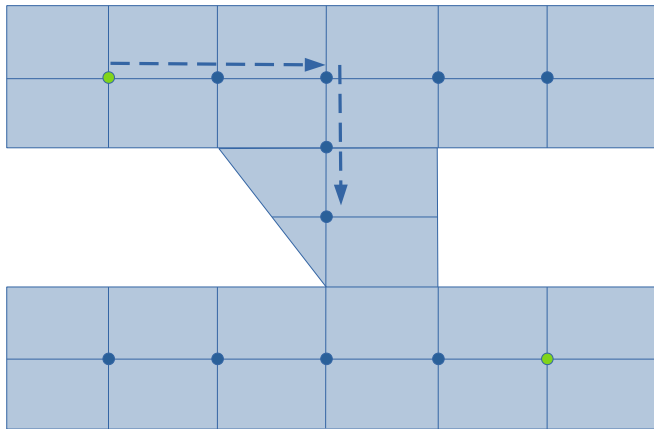
inner-point row of A :
$$\left[\cdots -\frac{1}{h^2} \cdots -\frac{1}{h^2} \frac{4}{h^2} -\frac{1}{h^2} \cdots -\frac{1}{h^2} \cdots \right]$$

boundary-point row of A :
$$\left[\cdots -\frac{1}{h^2} \frac{4}{h^2} -\frac{1}{h^2} \cdots -\frac{1}{h^2} \cdots \right]$$

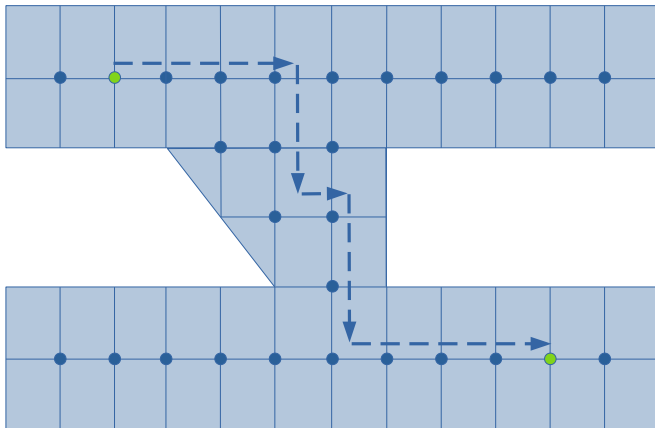
Weakly-chained property

A is called a weakly-chained matrix, if there is a direct walk from any inner point to any boundary point via inner edges only

A is not weakly-chained



A is weakly-chained



Discrete solution positivity

- ① If A is weakly-chained, then A is invertible, i.e. is an M matrix (invertible, positive definite Z matrix)
- ② If A is weakly-chained, then A^{-1} is positive, $A^{-1} \geq 0$, i.e., all its entries are positive (property of invertible M matrices)
- ③ Weakly-chained A is monotone, i.e., $A\mathbf{u} \geq \mathbf{0}$ implies $\mathbf{u} \geq \mathbf{0}$, discrete solution positivity property